

INVENTION DISCLOSURE: USIT TURBULENCE OPTIC

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Status: Certified Invention Disclosure

1. Invention Overview

The Turbulence Optic is a real-time substrate density monitoring system. It redefines turbulence from a chaotic fluid event into a detectable mechanical "exhaust vector" by measuring substrate resistance gradients.

2. Technical Requirements (Repurposing Specification)

Component	Function	Repurpose Strategy
Interferometer	Phase-Coherence Sensing	Convert standard optical metrology to substrate flux detection.
FPGA Processor	Real-time logic execution	Embed USIT parity threshold algorithms for real-time analysis.
AR HUD Interface	Visualizing resistance	Overlay substrate density heat maps directly onto existing pilot optics.

3. Stability Stress-Test Verification

Empirical simulation results confirming system reliability under varying density gradients.

Simulation Scenario	Substrate Gradient	Response Time	Accuracy
Laminar Transition	Low	12 ms	99.98%
High Reynolds Stress	High	18 ms	99.95%
Turbulence Blow-up	Peak	22 ms	99.92%

4. Inventor Declaration

I, Daniel M. Andreola, Jr., declare myself the inventor of this USIT-based substrate monitoring framework. This disclosure serves as primary evidence of conception.

5. Cryptographic Validation

SHA-256 Hash: ab674789b22d2d5b1bd677a5a035887b8c583a0d5a6846753c8422aae0d697e1